

Types of Numbers, HCF and LCM

Name: _____

IGCSE 0607 Extended Mathematics Practice Questions
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- 1 (a) Write the number two million, seven hundred thousand and fourteen in figures.

..... [1]

- (b) From the list of numbers below:

1, 4, 6, 10, 13, 21, 64

- (i) Write down a number that is both a square number and a cube number.

..... [1]

- (ii) Write down a triangle number greater than 5.

..... [1]

- (iii) Write down the reciprocal of $\frac{3}{5}$.

..... [1]

- (c) Express 120 as a product of its prime factors.

..... [2]

- (d) Find the highest common factor (HCF) and lowest common multiple (LCM) of 40 and 50.

- (i) Find the HCF of 40 and 50.

..... [1]

(ii) Find the LCM of 40 and 50.

..... [2]

2 (a) Write the number forty-two billion seven hundred million and four in figures.

..... [1]

(b) From the following list of numbers:

$\sqrt{16}$, 27, $\frac{2}{3}$, π , 28, 45

identify:

(i) the cube number,

..... [1]

(ii) the triangle number,

..... [1]

(iii) the irrational number.

..... [1]

3 (a) Write the number 30 040 008 in words using British standard formatting.

..... [1]

(b) Find the Highest Common Factor (HCF) of 45 and 105.

..... [2]

4 (a) Write the number 4002050 in words.

..... [1]

(b) From the list below:

$$\sqrt{9}, \quad \sqrt{10}, \quad \pi, \quad -\frac{4}{5}, \quad 0.75$$

(i) Write down an integer.

..... [1]

(ii) Write down an irrational number.

..... [1]

(c) Express 72 as a product of its prime factors in index form.

..... [2]

(d) Given that $X = 2^3 \times 3 \times 7$ and $Y = 2^2 \times 3^2 \times 5$:

(i) Find the HCF of X and Y .

..... [1]

(ii) Find the LCM of X and Y .

..... [2]

5 (a) Express 72 as a product of its prime factors. Give your answer in index form.

..... [2]

(b) Write down the reciprocal of 72.

..... [1]

6 Two numbers are given as $A = 2^2 \times 3 \times 5$ and $B = 2 \times 3^2 \times 7$.

(a) Find the Highest Common Factor (HCF) of A and B .

..... [2]

(b) Find the Lowest Common Multiple (LCM) of A and B . Give your answer as an integer.

..... [2]

7 A security light flashes every 42 seconds. A second security light flashes every 60 seconds. Both lights flash together at 19:30.

(a) Find the Lowest Common Multiple (LCM) of 42 and 60.

..... [2]

(b) Determine the next time both lights will flash together.

..... [2]

8 (a) Write the number three million, forty thousand and seven in figures.

..... [1]

(b) State whether each of the following statements is True or False.

(i) 17 is a prime number.

..... [1]

(ii) $\frac{3}{8}$ is an irrational number.

..... [1]

(iii) 0 is a natural number.

..... [1]

(c) Express 90 as a product of its prime factors.

..... [2]

(d) Find the highest common factor (HCF) and lowest common multiple (LCM) of 30 and 45.

(i) Find the HCF of 30 and 45.

..... [1]

(ii) Find the LCM of 30 and 45.

..... [2]